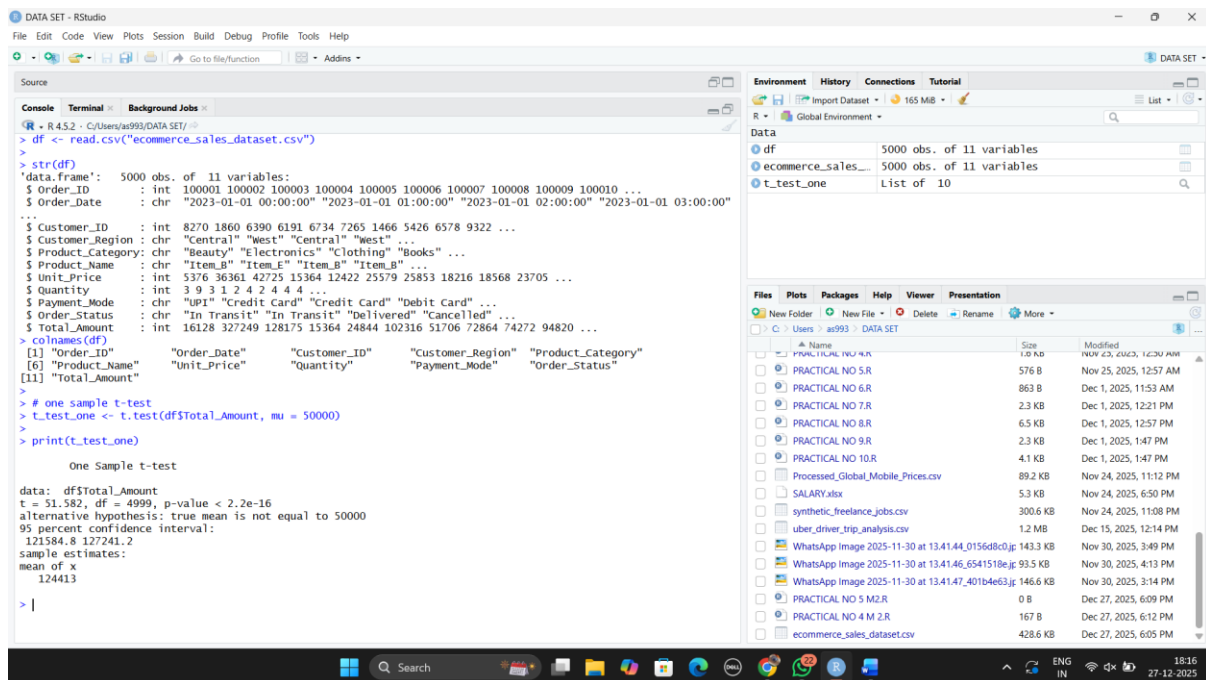


PRACTICAL NO 4 MODULE 2

AIM: Performing one-sample t-tests using t.test()

OUTPUT:-



```
R - R 4.5.2 : C:/Users/as993/DATA SET //
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source
Console Terminal Background Jobs
> df <- read.csv("ecommerce_sales_dataset.csv")
> str(df)
'data.frame': 5000 obs. of 11 variables:
 $ Order_ID : int 100001 100002 100003 100004 100005 100006 100007 100008 100009 100010 ...
 $ Order_Date : chr "2023-01-01 00:00:00" "2023-01-01 01:00:00" "2023-01-01 02:00:00" "2023-01-01 03:00:00"
 ...
 $ Customer_ID : int 8270 1860 6390 6191 6734 7265 1466 5426 6578 9322 ...
 $ Customer_Region : chr "Central" "West" "Central" "West" ...
 $ Product_Category : chr "Beauty" "Electronics" "Clothing" "Books" ...
 $ Product_Name : chr "Item_B" "Item_E" "Item_B" "Item_B" ...
 $ Unit_Price : int 5376 36361 42725 15364 12422 25579 25853 18216 18568 23705 ...
 $ Quantity : int 3 9 3 1 2 4 2 4 4 4 ...
 $ Payment_Mode : chr "UPI" "Credit Card" "Credit Card" "Debit Card" ...
 $ Order_Status : chr "In Transit" "In Transit" "Delivered" "Cancelled" ...
 $ Total_Amount : int 16128 327249 128175 15364 24844 102316 51706 72864 74272 94820 ...
> colnames(df)
[1] "Order_ID" "Order_Date" "Customer_ID" "Customer_Region" "Product_Category"
[6] "Product_Name" "Unit_Price" "Quantity" "Payment_Mode" "Order_Status"
[11] "Total_Amount"
>
> # one sample t-test
> t_test_one <- t.test(df$Total_Amount, mu = 50000)
> print(t_test_one)

One Sample t-test

data: df$Total_Amount
t = 51.582, df = 4999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 50000
95 percent confidence interval:
 121584.8 127241.2
sample estimates:
mean of x
 124413
> |
```

The screenshot shows the RStudio interface with the console output of a one-sample t-test. The data is loaded from 'ecommerce_sales_dataset.csv'. The t-test is performed on the 'Total_Amount' variable against a null hypothesis mean of 50,000. The output shows a t-statistic of 51.582, degrees of freedom of 4999, and a p-value less than 2.2e-16, indicating a highly significant result. The 95% confidence interval for the mean is approximately 121,584.8 to 127,241.2.